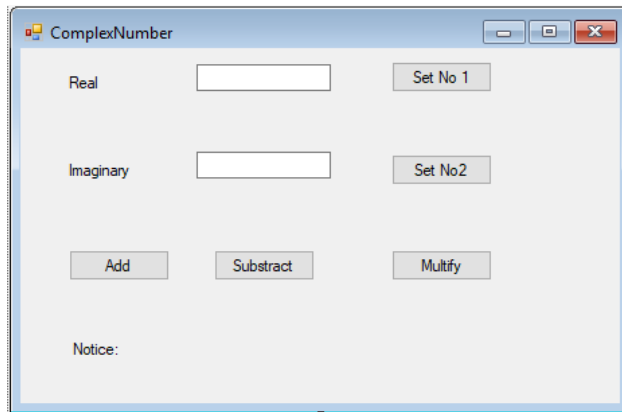


(Ch2)

Ôn Tập

Note: D:\HK1_2018\OOP\OOPC#\day5\Confuse



```
public partial class Form2 : Form
{
    public Form2()
    {
        InitializeComponent();
    }
    private ComplexNumber x = new ComplexNumber();
    private ComplexNumber y = new ComplexNumber();

    private void firstButton_Click(object sender, EventArgs e)
    {
        x.Real = Int32.Parse(realTextBox.Text);
        x.Imaginary = Int32.Parse(imaginaryTextBox.Text);
        realTextBox.Clear();
        imaginaryTextBox.Clear();
        statusLabel.Text = "First Complex Number is: " + x;
    }

    private void secondButton_Click(object sender, EventArgs e)
    {
        y.Real = Int32.Parse(realTextBox.Text);
        y.Imaginary = Int32.Parse(imaginaryTextBox.Text);
        realTextBox.Clear();
        imaginaryTextBox.Clear();
        statusLabel.Text = "Second Complex Number is: " + y;
    }

    private void addButton_Click(object sender, EventArgs e)
    {
        statusLabel.Text = x + " + " + y + " = " + (x + y);
    }

    private void subtractButton_Click(object sender, EventArgs e)
    {
        statusLabel.Text = x + " - " + y + " = " + (x - y);
    }

    private void multiplyButton_Click(object sender, EventArgs e)
    {
        statusLabel.Text = x + " * " + y + " = " + (x * y);
    }
}
```

- ComplexNumber.cs

```

class ComplexNumber
{
    private int real;
    private int imaginary;

    // default constructor
    public ComplexNumber() { }

    // constructor
    public ComplexNumber(int a, int b)
    {
        Real = a;
        Imaginary = b;
    }

    // return string representation of ComplexNumber
    public override string ToString()
    {
        return "(" + real +
            (imaginary < 0 ? " - " + (imaginary * -1) :
             " + " + imaginary) + "i )";
    }

    // property Real
    public int Real
    {
        get { return real;}
        set{ real = value;}
    } // end property Real

```

```

// property Imaginary
public int Imaginary
{
    get{return imaginary;}
    set{ imaginary = value;}
} // end property Imaginary

// overload the addition operator
public static ComplexNumber operator +(
    ComplexNumber x, ComplexNumber y)
{
    return new ComplexNumber(
        x.Real + y.Real, x.Imaginary + y.Imaginary);
}
public static ComplexNumber Add(ComplexNumber x,
    ComplexNumber y)
{
    return x + y;
}

// overload the subtraction operator
public static ComplexNumber operator -(
    ComplexNumber x, ComplexNumber y)
{
    return new ComplexNumber(
        x.Real - y.Real, x.Imaginary - y.Imaginary);
}

```

```

// provide alternative to overloaded - operator
// for subtraction
public static ComplexNumber Subtract(ComplexNumber x,
    ComplexNumber y)
{
    return x - y;
}

// overload the multiplication operator
public static ComplexNumber operator *(
    ComplexNumber x, ComplexNumber y)
{
    return new ComplexNumber(
        x.Real * y.Real - x.Imaginary * y.Imaginary,
        x.Real * y.Imaginary + y.Real * x.Imaginary);
}

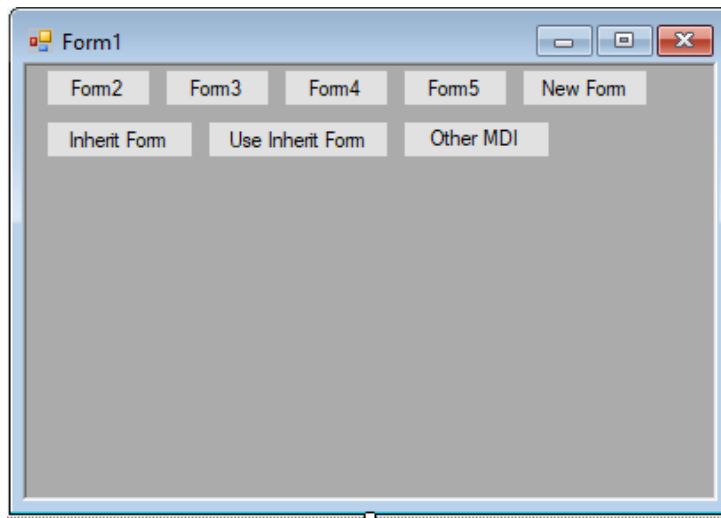
public static ComplexNumber Multiply(ComplexNumber x,
    ComplexNumber y)
{
    return x * y;
}
}

```

Form và các định dạng form

1. MDI Form: form được gọi là MDI form khi properties *IsMdiContainer* được đặt là TRUE

Thiết kế Form1 như sau:



Code chức năng:

```

private void button1_Click(object sender, EventArgs e)
{
    Form frm = new Form2();
    frm.IsMdiContainer = true;
    frm.Show ();
}

private void button2_Click(object sender, EventArgs e)
{
    // khai bao va tao form
    Form frm = new Form3();
    // MDI form la form hien hanh
    frm.MdiParent = this;
    // dua form moi len man hinh
    frm.Show();
}

private void button4_Click(object sender, EventArgs e)
{
    Form frm = new Form5();
    frm.ShowDialog();
}

private void button5_Click(object sender, EventArgs e)
{
    Form frm = new Form();
    frm.Text = "New Form";
    frm.Name = "frmNewForm";
    frm.Click += delegate(object sender1, EventArgs e1)
    { System.Windows.Forms.MessageBox.Show("Click on the new Form!", "Vi du Form"); };

    frm.Show();
}

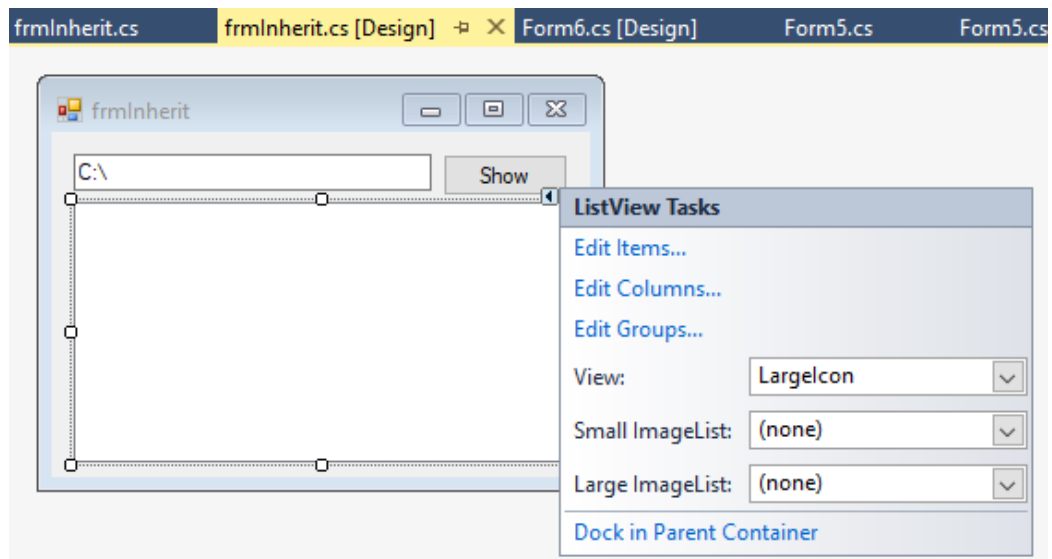
private void button6_Click(object sender, EventArgs e)
{
    Form frm = new frmInherit ();
    frm.ShowDialog();
}

private void button7_Click(object sender, EventArgs e)
{
    Form frm = new Form6();
    frm.ShowDialog();
}

private void button8_Click(object sender, EventArgs e)
{
    Form frm = new Form5();
    frm.MdiParent = Form2.ActiveForm ;
    frm.Show();
}

```

- Chú ý nút New Form: là delegate mà các event thường dùng
- Inherit Form:
Thiết kế form như sau, trong đó có thêm control ListView

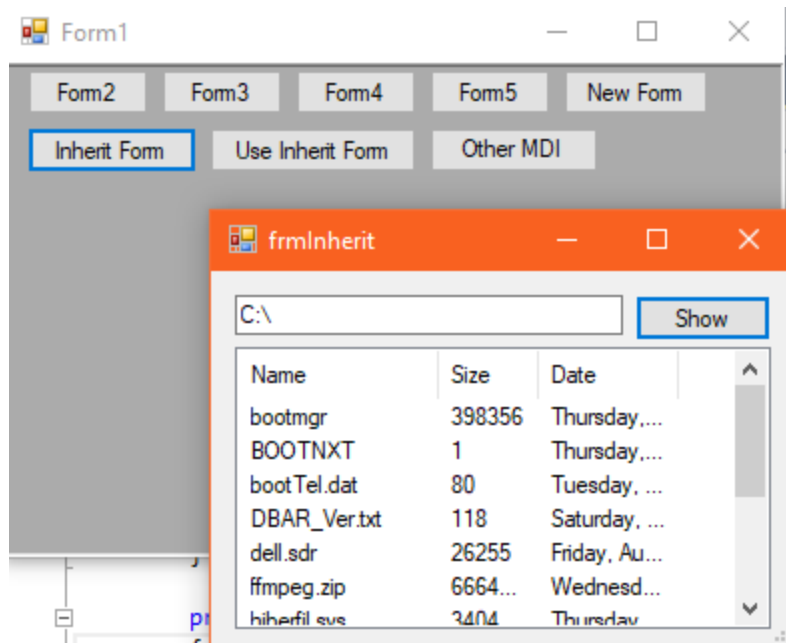


Code:

```
public partial class frmInherit : Form
{
    public frmInherit()
    {
        InitializeComponent();
    }

    private void btnShow_Click(object sender, EventArgs e)
    {
        try
        {
            DirectoryInfo dir = new DirectoryInfo(txtPath.Text);
            lvData.Columns.Add("Name", 100, 0);
            lvData.Columns.Add("Size", 50, 0);
            lvData.Columns.Add("Date", 70, 0);
            lvData.View = View.Details;
            ListViewItem item1;
            foreach(FileInfo d in dir.GetFiles("*.txt"))
            {
                item1 = new ListViewItem( d.Name);
                item1.SubItems.Add(d.Length.ToString());
                item1.SubItems.Add(d.CreationTime.ToLongDateString());
                lvData.Items.Add(item1);
            }
        }
        catch(Exception ex)
        {
            MessageBox.Show (ex.Message );
        }
    }
}
```

Kết quả:



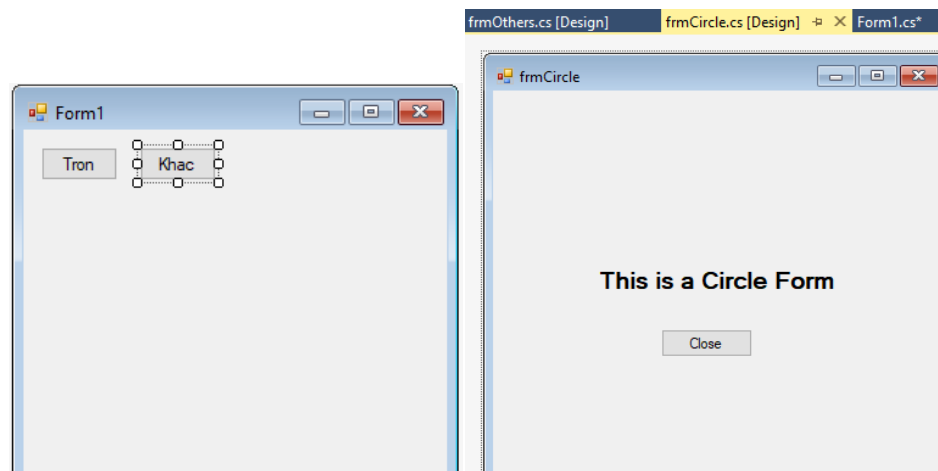
Sinh viên có thể thay đổi C hay D, E.

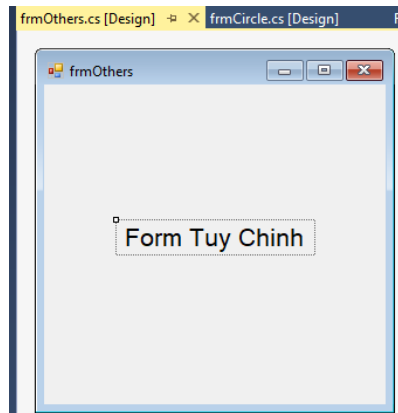
- Use InheritForm chỉ là nút gọi Inherit Form (1 form khác, kế thừa từ Inherit Form), trong code nó là 1 Normal Form (button7)

```
public partial class Form6 : frmInherit
{
    public Form6()
    {
        InitializeComponent();
    }
}
```

Các hình dạng của Form

- Thiết kế 3 form
- Form 1 là form chính





Code:

```
public partial class Form1 : Form
{
    public Form1()
    {
        InitializeComponent();
    }

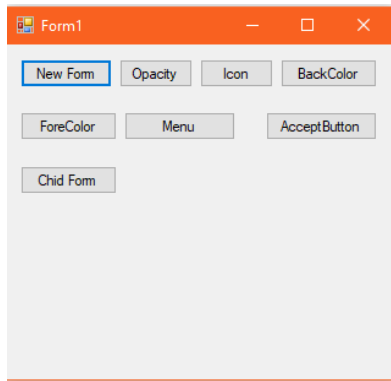
    private void button1_Click(object sender, EventArgs e)
    {
        Form frm = new frmCircle();
        GraphicsPath shape;
        shape = new GraphicsPath();
        shape.AddEllipse(15, 25, this.Width, this.Height);
        frm.Region = new Region(shape);
        frm.Show();
    }

    private void button2_Click(object sender, EventArgs e)
    {
        Form frm = new frmOthers();
        GraphicsPath shape;
        shape = new GraphicsPath();
        Point point1 = new Point(100, 50);
        Point point2 = new Point(250, 100);
        Point point3 = new Point(300, 200);
        Point point4 = new Point(180, 300);
        Point point5 = new Point(80, 200);
        Point point6 = new Point(30, 150);
        Point[] points = { point1, point2, point3, point4, point5, point6 };

        shape.AddCurve(points);
        frm.Region = new Region(shape);
        frm.Show();
    }
}
```

Các thuộc tính cơ bản của Form

- Thiết kế Form 1 bao gồm các nút chức năng



- Nút New Form

Tạo ra 1 form mới bằng code

```
private void button1_Click(object sender, EventArgs e)
{
    Form frm = new Form();
    frm.Text = "New Form";
    frm.Name = "frmNewForm";
    frm.Show();
}
```

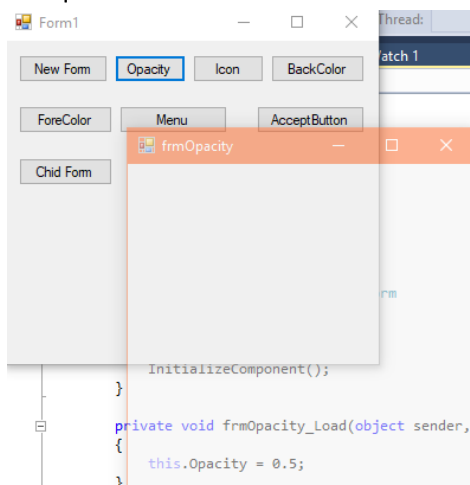
- Nút Opacity: thuộc tính làm trong suốt Form

Nút này gọi form được vài đặt thuộc tính opacity như sau:

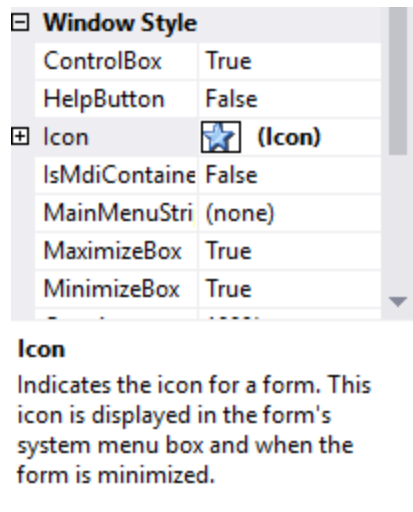
```
public partial class frmOpacity : Form
{
    public frmOpacity()
    {
        InitializeComponent();
    }

    private void frmOpacity_Load(object sender, EventArgs e)
    {
        this.Opacity = 0.5;
    }
}
```

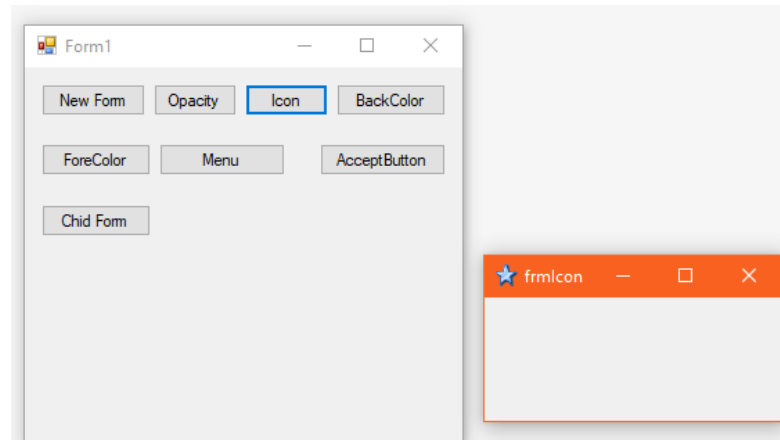
Kết quả:



- Nút Icon : gọi 1 form mà cài đặt thuộc tính icon (có icon đính kèm với Form)



Kết quả:



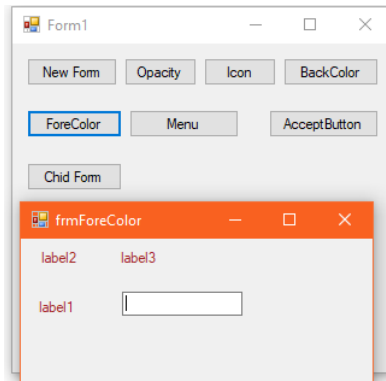
- Nút Back và Fore Color: cài đặt màu trước và sau cho Form

Gọi 2 form có thay đổi màu

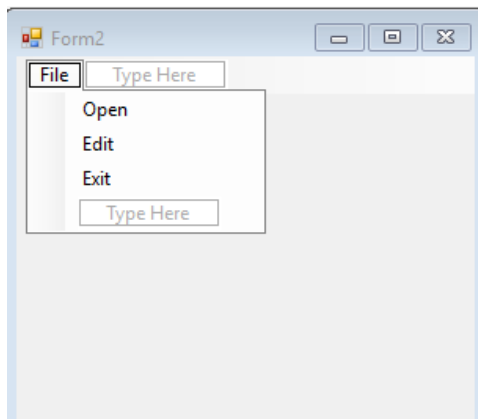
```
private void frmForeColor_Load(object sender, EventArgs e)
{
    this.ForeColor = Color.Brown;
}

private void frmBackColor_Load(object sender, EventArgs e)
{
    this.BackColor = Color.Azure;
}
```

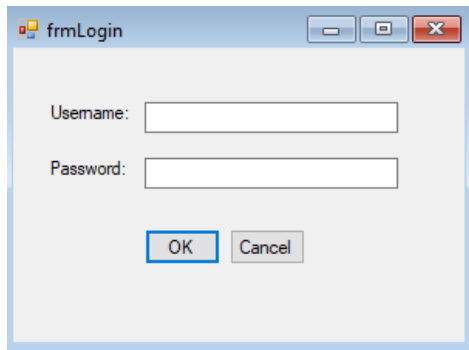
Kết quả:



- Nút Menu gọi đến Form có chứa 1 ToolStripMenu đơn giản



- Nút AcceptButton gọi đến Form Login: (Phần này sẽ làm chi tiết sau)



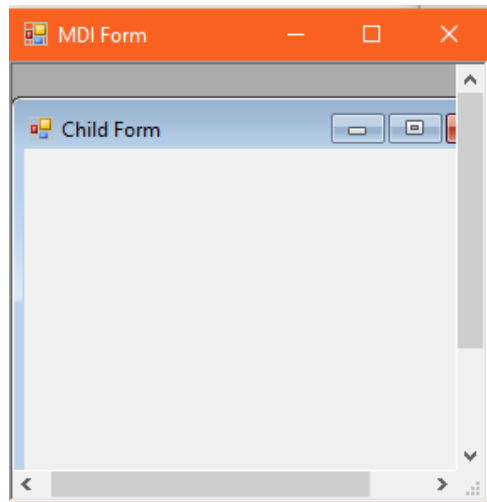
```

private void btnOK_Click(object sender, EventArgs e)
{
    if (txtUser.Text == "")
    {
        MessageBox.Show("Enter Username");
        txtUser.Focus();
        return;
    }
    if (txtPwd.Text == "")
    {
        MessageBox.Show("Enter Password");
        txtPwd.Focus();
        return;
    }
}

private void btnCancel_Click(object sender, EventArgs e)
{
    Application.Exit();
}

```

- Nút Child Form, sinh viên làm và giải thích code



Code:

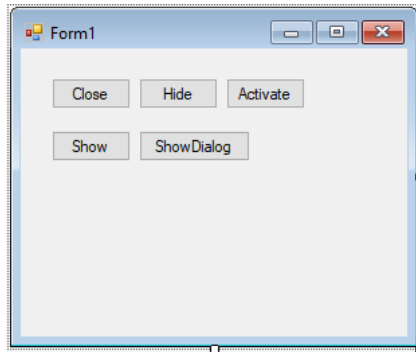
```

private void button8_Click(object sender, EventArgs e)
{
    Form frm = new Form();
    frm.Text = "MDI Form";
    frm.IsMdiContainer = true;
    frm.Load += delegate(object sender1, EventArgs e1)
    {
        Form ChildForm = new Form();
        ChildForm.Text = "Child Form";
        ChildForm.MdiParent = frm;
        ChildForm.WindowState =
            FormWindowState.Maximized;
        ChildForm.Show();
    };
    frm.ShowDialog();
}

```

Method cơ bản của Form

- Thiết kế Form và có code như sau:



```
public partial class Form1 : Form
{
    public Form1()
    {
        InitializeComponent();
    }

    private void button1_Click(object sender, EventArgs e)
    {
        this.Close();
    }

    private void button2_Click(object sender, EventArgs e)
    {
        this.Hide();
    }

    private void button3_Click(object sender, EventArgs e)
    {
        Form frm = new frmActivate();
        frm.ShowDialog();
    }

    private void button4_Click(object sender, EventArgs e)
    {
        Form frm = new frmActivate();
        frm.Show();
    }

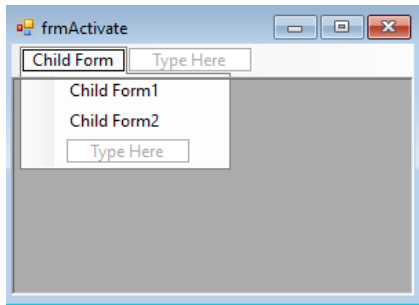
    private void button5_Click(object sender, EventArgs e)
    {
        Form frm = new frmActivate();
        frm.ShowDialog();
    }
}
```

- Ẩn Form (Hide): thử đoạn code sau bằng cách thêm chức năng của đoạn code vào Form 1

```
private void btt1_Click(object sender, EventArgs e)
{
    Form2 frm2 = new Form2();
    this.Hide();
    frm2.Show();
}

private void btt2_Click(object sender, EventArgs e)
{
    Form1 frm1 = new Form1();
    this.Close();
    frm1.Show();
}
```

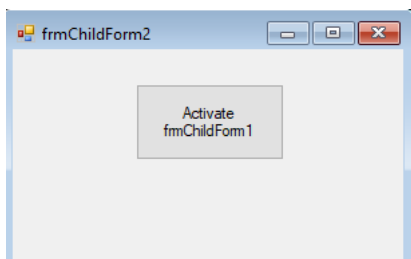
- Active:



```
public partial class frmActivate : Form
{
    public frmActivate()
    {
        InitializeComponent();
    }

    private void childForm1ToolStripMenuItem_Click(object sender, EventArgs e)
    {
        Form frm = new frmChildForm1();
        frm.MdiParent = this;
        frm.Show();
    }

    private void childForm2ToolStripMenuItem_Click(object sender, EventArgs e)
    {
        Form frm = new frmChildForm2();
        frm.MdiParent = this;
        frm.Show();
    }
}
```



```
public partial class frmChildForm2 : Form
{
    public frmChildForm2()
    {
        InitializeComponent();
    }

    private void button1_Click(object sender, EventArgs e)
    {
        Form frm = frmActivate.ActiveForm;
        foreach (Form f in frm.MdiChildren)
        {
            if (f.Text == "frmChildForm1")
                f.Activate();
        }
    }
}
```

ChildForm1 chỉ là Form rỗng, Sinh viên tự làm.

- Nút Show: so sánh Show() và ShowDialog()?

Event của Form:

Thiết kế Form1 với design và code sau:

```

private void button1_Click(object sender, EventArgs e)
{
    Form frm = new frmClosed();
    frm.ShowDialog();
}

private void button2_Click(object sender, EventArgs e)
{
    Form frm = new frmClosing();
    frm.ShowDialog();
}

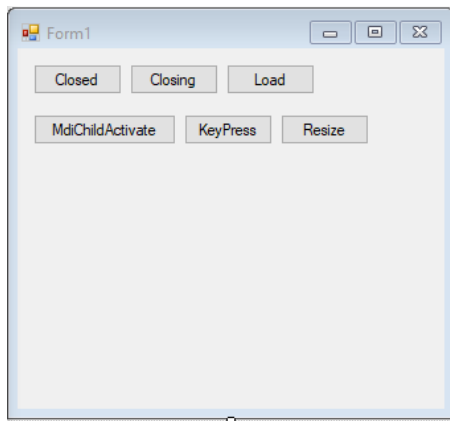
private void button3_Click(object sender, EventArgs e)
{
    Form frm = new frmLoad();
    frm.ShowDialog();
}

private void button4_Click(object sender, EventArgs e)
{
    Form frm = new frmMDIForm();
    frm.ShowDialog();
}

private void button5_Click(object sender, EventArgs e)
{
    Form frm = new frmKeyPress();
    frm.ShowDialog();
}

private void button6_Click(object sender, EventArgs e)
{
    Form frm = new frmResize();
    frm.ShowDialog();
}

```



- Closed:

```

public partial class frmClosed : Form
{
    public frmClosed()
    {
        InitializeComponent();
    }

    private void frmClosed_FormClosed(
        object sender, FormClosedEventArgs e)
    {
        MessageBox.Show(e.CloseReason.ToString());
    }
}

```

- Closing:

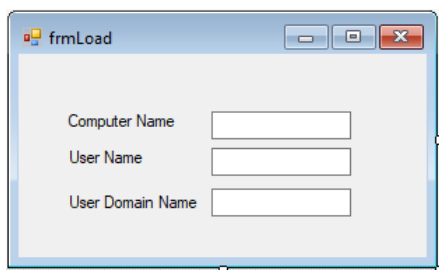
```

public partial class frmClosed : Form
{
    public frmClosed()
    {
        InitializeComponent();
    }

    private void frmClosed_FormClosed(
        object sender, FormClosedEventArgs e)
    {
        MessageBox.Show(e.CloseReason.ToString());
    }
}

```

- Load:



```

public partial class frmLoad : Form
{
    public frmLoad()
    {
        InitializeComponent();
    }

    private void frmLoad_Load(object sender, EventArgs e)
    {
        textBox1.Text = SystemInformation.ComputerName;
        textBox2.Text = SystemInformation.UserName;
        textBox3.Text = SystemInformation.UserDomainName;
    }
}

```

- MdiChildActive:


```

public partial class frmMDIForm : Form
{
    public frmMDIForm()
    {
        InitializeComponent();
    }

    private void child1ToolStripMenuItem_Click(object sender, EventArgs e)
    {
        Form frm = new Child1();
        frm.MdiParent = this;
        frm.Show();
    }

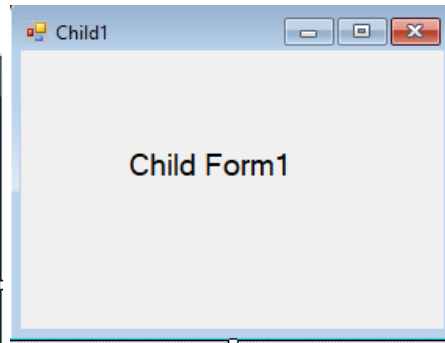
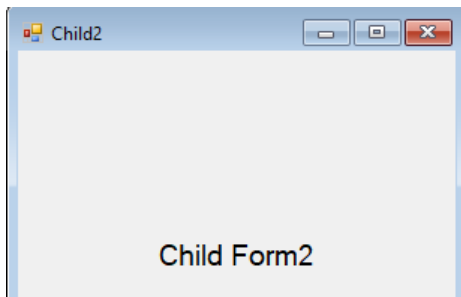
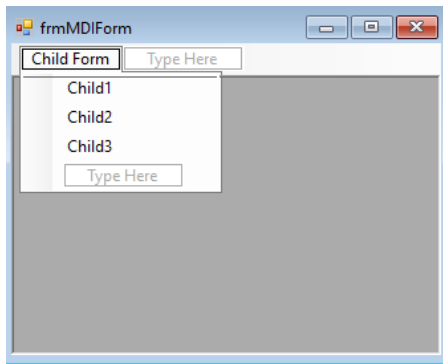
    private void child2ToolStripMenuItem_Click(object sender, EventArgs e)
    {
        Form frm = new Child2();
        frm.MdiParent = this;
        frm.Show();
    }

    private void child3ToolStripMenuItem_Click(object sender, EventArgs e)
    {
        Form frm = new Child3();
        frm.MdiParent = this;
        frm.Show();
    }

    private void childFormToolStripMenuItem_Click(object sender, EventArgs e)
    {
    }

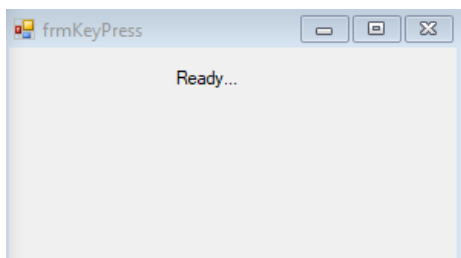
    private void frmMDIForm_MdiChildActivate(object sender, EventArgs e)
    {
        foreach (Form frm in this.MdiChildren)
        {
            label1.Text = frm.Name;
        }
    }
}

```



Tương tự Child3

- Key Press:



```

public partial class frmKeyPress : Form
{
    public frmKeyPress()
    {
        InitializeComponent();
    }

    private void frmKeyPress_KeyPress(object sender, KeyPressEventArgs e)
    {
        label1.Text = "Pressed by key : "
            + e.KeyChar.ToString();
    }
}

```

- Resize:

```

public partial class frmResize : Form
{
    public frmResize()
    {
        InitializeComponent();
    }

    private void frmResize_Resize(object sender, EventArgs e)
    {
        richTextBox1.Width = this.Width - 30;
        richTextBox1.Height = this.Height - 50;
    }
}

```

Activated Form:

“When the application is active and has multiple forms, the active form is the form with the input focus. A form that is not visible cannot be the active form. The simplest way to activate a visible form is to click it or use an appropriate keyboard combination”

Chu trình của Form:

Thứ tự xuất hiện các event khi startup 1 form or control

```

Control.HandleCreated
Control.BindingContextChanged
Form.Load
Control.VisibleChanged
Form.Activated
Form.Shown

```

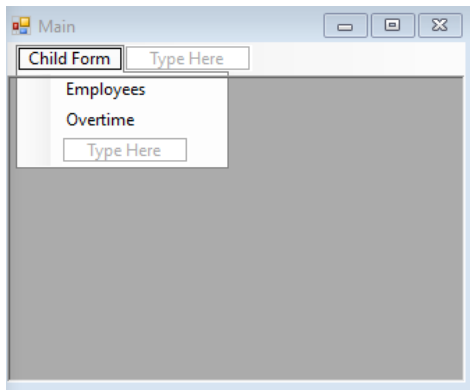
Thứ tự xuất hiện các event khi exit 1 form or control

```

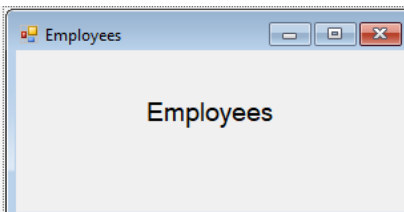
Form.Closing
Form.FormClosing
Form.Closed
Form.FormClosed
Form.Deactivate

```

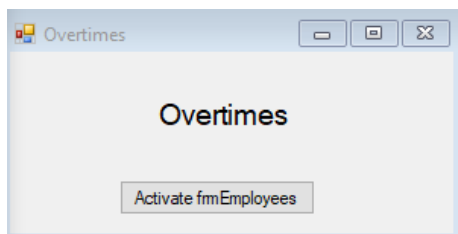
- Thiết kế Main Form với Menu Strip control như sau:



- Employees:



- Overtime:



⇒ Thực thi và giải thích?

```
public partial class frmMain : Form
{
    public frmMain()
    {
        InitializeComponent();
    }

    private void employeesToolStripMenuItem_Click(object sender, EventArgs e)
    {
        Form frm = new frmEmployees();
        frm.MdiParent = this;

        frm.Show();
    }

    private void overtimeToolStripMenuItem_Click(object sender, EventArgs e)
    {
        Form frm = new frmOvertimes();
        frm.MdiParent = this;
        frm.Show();
    }
}
```

```
public partial class frmEmployees : Form
{
    public frmEmployees()
    {
        InitializeComponent();
    }
    int i = 0;
    private void frmEmployees_Activated(object sender, EventArgs e)
    {
        i += 1;
        this.label2.Text = "Activated " + i.ToString();
    }
}
```

```
public partial class frmOvertimes : Form
{
    public frmOvertimes()
    {
        InitializeComponent();
    }

    private void button1_Click(object sender, EventArgs e)
    {
        Form frm = frmMain.ActiveForm;
        foreach (Form f in frm.MdiChildren)
            if (f.Name == "frmEmployees")
                f.Activate();
    }
}
```

Quản lý Sinh Viên

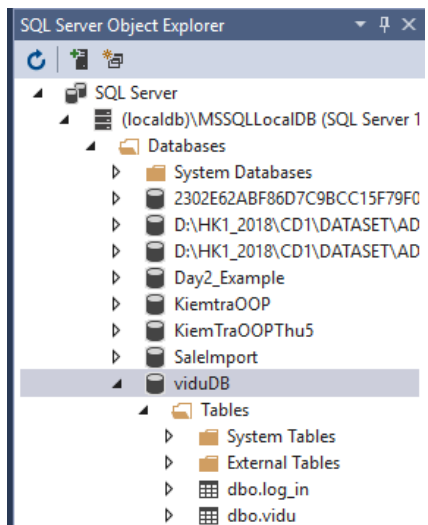
Class Diagram



Database

Tạo database chứa các tables liên quan đến Project.

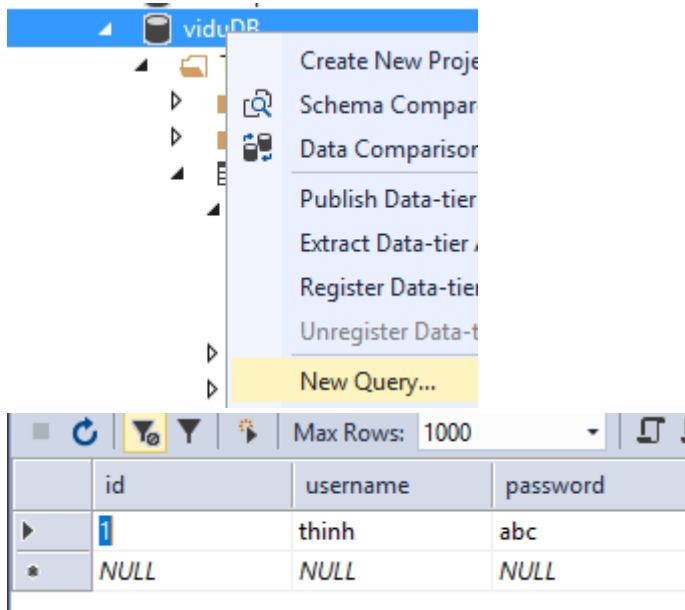
1. Sinh viên tạo database chứa table log_in như hình



2. Table có các cột

	id	username	password
*	NULL	NULL	NULL

Table chứa username và password để login vào chương trình. Sinh viên có thể dùng lệnh query insert into để thêm data vào và data này dùng để điền vào textbox trong Form Login. Ví dụ:



MY_DB

- File quản lý kết nối database
- Code gợi ý:

```

SqlConnection con = new SqlConnection("Data Source :

// get the connection
public SqlConnection getConnection
{
    get
    {
        return con;
    }
}

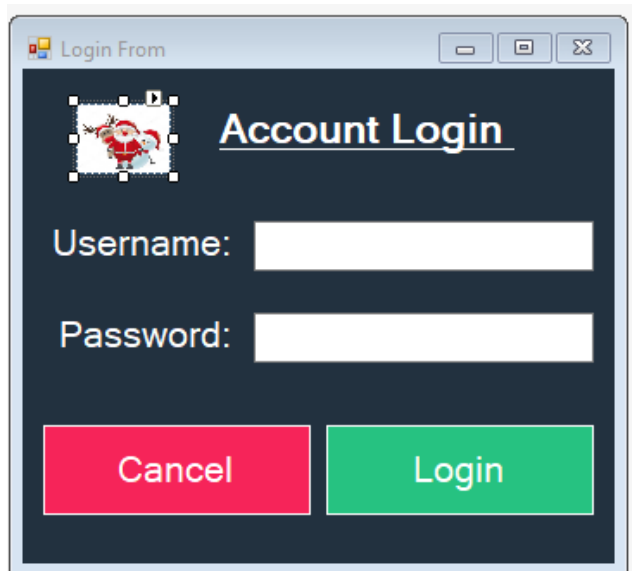
// open the connection
public void openConnection()
{
    if ((con.State == ConnectionState.Closed))
    {
        con.Open();
    }
}

// close the connection
public void closeConnection()
{
    if ((con.State == ConnectionState.Open))
    {
        con.Close();
    }
}

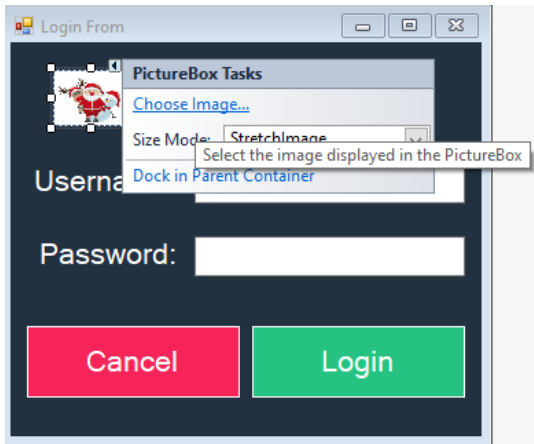
```

Login Form.cs

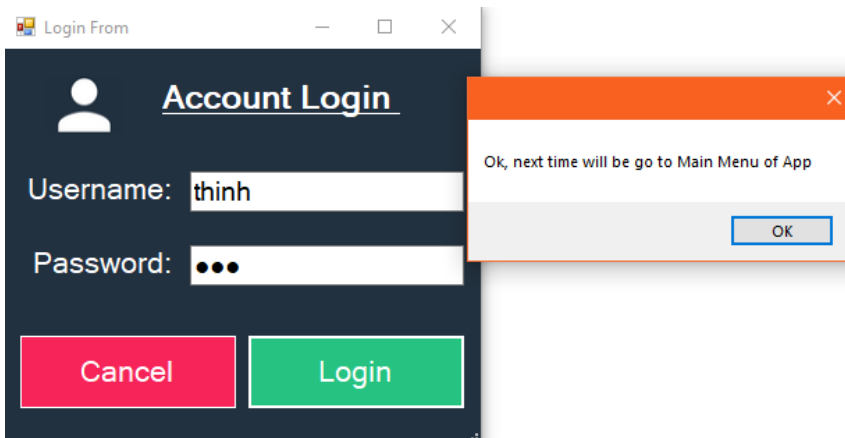
Thiết kế form như sau:



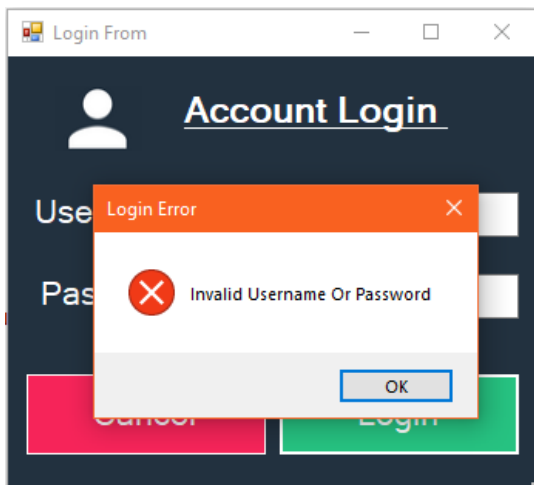
- Control PictureBox dùng để đưa hình vào Form



- Nhập đúng user va password thì thông báo:

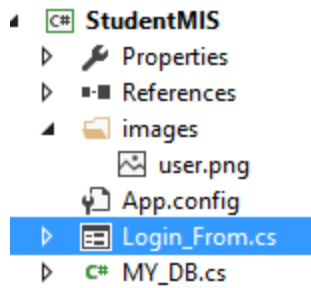


- Nhập sai sẽ báo :



Gợi ý:

Đối với PicutureBox, sinh viên có thể tạo Folder ;



Và có thể dùng code gợi ý sau:

```
private void Login_From_Load(object sender, EventArgs e)
{
    // đưa hình tu folder
    pictureBox1.Image = Image.FromFile("../images/user.png");
}
```

Sinh viên so sánh dữ liệu nhập vào từ textbox với dữ liệu trong database sẽ hoàn thành phần này. Sinh viên có thể dùng cách viết Sql hay Linq đều được, ví dụ dùng sqlCommand:

```
SqlCommand command = new SqlCommand("SELECT * FROM log_in WHERE username = @User AND password = @Pass", db.getConnection);

command.Parameters.Add("@User", SqlDbType.VarChar).Value = TextBoxUsername.Text;
command.Parameters.Add("@Pass", SqlDbType.VarChar).Value = TextBoxPassword.Text;
```

Trong đó db là đối tượng lớp My_DB bên trên.

Parameter @ được giải thích trong file khác.

Phần mở rộng:

Trong Day02 sau khi học xong cách thêm dữ liệu vào bảng. Sinh viên làm các chức năng trong hình

